D-ANMOL IT SOLUTIONS

# Artificial Intelligence & Machine

Learning

### 3 Months | Foundation Track

A step-by-step practical learning roadmap with explained tools, applied examples, project milestones and career preparation.

## 01 WHAT CAN YOU BUILD AFTER THIS COURSE?

- Student Performance Predictor: Train a model to estimate student scores from attendance and study patterns.
- Spam Email Classification: Classify text messages as spam or legitimate with a tested pipeline.

### Possible career directions

ML trainee, junior AI/ML developer, data analyst and applied AI project intern. Roles are illustrative and depend on skills, experience and hiring criteria.

## Your learning journey

### From learning to practical delivery

| Learn concepts | Practise skills | Build projects | Show portfolio |
|---|---|---|---|

## 02 STEP-BY-STEP COURSE SYLLABUS

Modules are ordered from foundations to application. Each includes what you learn, why it matters and a practical classroom activity.

### STEP 01 | AI, ML & Deep Learning Overview

What you will study:

- AI history: understand the concept, practise it with a guided example, and apply it in a relevant task.
- narrow AI: understand the concept, practise it with a guided example, and apply it in a relevant task.
- ML types: understand the concept, practise it with a guided example, and apply it in a relevant task.
- deep learning: understand the concept, practise it with a guided example, and apply it in a relevant task.
- use cases: understand the concept, practise it with a guided example, and apply it in a relevant task.
- workflow and responsible AI: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: It provides a foundation for later modules and real project implementation.

Hands-on example: Map a real-world problem to a suitable AI approach.

## STEP 02 | Python for AI

What you will study:

• Variables: understand the concept, practise it with a guided example, and apply it in a relevant task.

• data types: understand the concept, practise it with a guided example, and apply it in a relevant task.

• conditions: understand the concept, practise it with a guided example, and apply it in a relevant task.

• loops: understand the concept, practise it with a guided example, and apply it in a relevant task.

• functions: understand the concept, practise it with a guided example, and apply it in a relevant task.

• collections: understand the concept, practise it with a guided example, and apply it in a relevant task.

• modules and file handling: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: It provides a foundation for later modules and real project implementation.

Hands-on example: Write reusable Python programs for data tasks. D-ANMOL IT SOLUTIONS | COURSE CURRICULUM Page 2

## STEP 03 | NumPy & Pandas

What you will study:

• Arrays: understand the concept, practise it with a guided example, and apply it in a relevant task.

• indexing: understand the concept, practise it with a guided example, and apply it in a relevant task.

• DataFrames: understand the concept, practise it with a guided example, and apply it in a relevant task.

• importing data: understand the concept, practise it with a guided example, and apply it in a relevant task.

• filtering: understand the concept, practise it with a guided example, and apply it in a relevant task.

• missing values and transformations: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: It provides a foundation for later modules and real project implementation.

Hands-on example: Clean a raw dataset and produce an analysis-ready table.

## STEP 04 | Maths & Statistics for ML

What you will study:

• Descriptive statistics: understand the concept, practise it with a guided example, and apply it in a relevant task.

• probability: understand the concept, practise it with a guided example, and apply it in a relevant task.

• distributions: understand the concept, practise it with a guided example, and apply it in a relevant task.

• vectors: understand the concept, practise it with a guided example, and apply it in a relevant task.

• matrices and correlation: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: You use this skill to solve practical problems, improve accuracy and build reliable outputs.

Hands-on example: Summarise a dataset and interpret statistical relationships.

## STEP 05 | Data Visualisation & EDA

What you will study:

• Matplotlib: understand the concept, practise it with a guided example, and apply it in a relevant task.

• Seaborn: understand the concept, practise it with a guided example, and apply it in a relevant task.

• distributions: understand the concept, practise it with a guided example, and apply it in a relevant task.

• outliers: understand the concept, practise it with a guided example, and apply it in a relevant task.

• relationships and exploratory analysis: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: You use this skill to solve practical problems, improve accuracy and build reliable outputs.

Hands-on example: Create an EDA notebook with visual findings.

## STEP 06 | ML Workflow & Feature Engineering

What you will study:

• Problem framing: understand the concept, practise it with a guided example, and apply it in a relevant task.

• labels: understand the concept, practise it with a guided example, and apply it in a relevant task.

• features: understand the concept, practise it with a guided example, and apply it in a relevant task.

• train/test split: understand the concept, practise it with a guided example, and apply it in a relevant task.

• encoding: understand the concept, practise it with a guided example, and apply it in a relevant task.

• scaling and leakage prevention: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: You use this skill to solve practical problems, improve accuracy and build reliable outputs.

Hands-on example: Prepare a reusable model-ready dataset.

## STEP 07 | Supervised Learning: Regression

What you will study:

• Linear regression: understand the concept, practise it with a guided example, and apply it in a relevant task.

• multiple regression: understand the concept, practise it with a guided example, and apply it in a relevant task.

• assumptions: understand the concept, practise it with a guided example, and apply it in a relevant task.

• residuals and regression metrics: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: You use this skill to solve practical problems, improve accuracy and build reliable outputs.

Hands-on example: Build a house-price or sales prediction model.

## STEP 08 | Supervised Learning: Classification

What you will study:

• Logistic regression: understand the concept, practise it with a guided example, and apply it in a relevant task.

• KNN: understand the concept, practise it with a guided example, and apply it in a relevant task.

• decision trees: understand the concept, practise it with a guided example, and apply it in a relevant task.

• confusion matrix: understand the concept, practise it with a guided example, and apply it in a relevant task.

• precision: understand the concept, practise it with a guided example, and apply it in a relevant task.

• recall and F1: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: You use this skill to solve practical problems, improve accuracy and build reliable outputs.

Hands-on example: Build a spam or customer-churn classifier.

### STEP 09 | Ensemble Models & Model Selection

What you will study:

• Random forest: understand the concept, practise it with a guided example, and apply it in a relevant task.

• boosting concepts: understand the concept, practise it with a guided example, and apply it in a relevant task.

• cross-validation and model comparison: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: You use this skill to solve practical problems, improve accuracy and build reliable outputs.

Hands-on example: Compare multiple models and justify the selected model.

### STEP 10 | Unsupervised Learning

What you will study:

• K-Means: understand the concept, practise it with a guided example, and apply it in a relevant task.

• hierarchical clustering: understand the concept, practise it with a guided example, and apply it in a relevant task.

• PCA concepts and segmentation use cases: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: You use this skill to solve practical problems, improve accuracy and build reliable outputs.

Hands-on example: Segment customers or products using clustering.

## 03 TOOLS, LIBRARIES & THEIR USES

This section explains where each tool or library fits in a practical project. Some advanced tools may be introduced as demonstrations in shorter tracks.

| Tool / library | What it is used for |
|---|---|
| NumPy | Perform fast numerical operations on arrays. |
| Pandas | Load, clean and transform tabular datasets. |
| Matplotlib / Seaborn | Visualize distributions, trends and model results. |

| Tool / library | What it is used for |
| --- | --- |
| Scikit-learn | Preprocess features, train models and evaluate predictions. |
| Jupyter Notebook | Run experiments and document analysis step by step. |

# 04 PROJECT-BASED LEARNING

Included: 2 project(s). Every project includes planning, implementation, testing and a presentation of results.

### NORMAL PROJECT 1: Student Performance Predictor

What you will build: Train a model to estimate student scores from attendance and study patterns.

Where tools are used: Pandas cleans records; Scikit-learn trains regression; Matplotlib plots error.

Final deliverable: Notebook, evaluation report and prediction demo.

### Project workflow

| Input / brief | Build & test | Review results | Present output |
| --- | --- | --- | --- |

Live project means an applied, end-to-end project based on a realistic business scenario or suitable dataset; access to an external client or production deployment is not implied.

### LIVE PROJECT 2: Spam Email Classification

What you will build: Classify text messages as spam or legitimate with a tested pipeline.

Where tools are used: Scikit-learn TF-IDF vectorizer and classifier; Pandas for labels.

Final deliverable: Classifier, confusion matrix and test samples.

### Project workflow

| Input / brief | Build & test | Review results | Present output |
| --- | --- | --- | --- |

Live project means an applied, end-to-end project based on a realistic business scenario or suitable dataset; access to an external client or production deployment is not implied.

# 05 TRAINING PLAN & ASSESSMENTS

Suggested duration: 3 Months (approximately 12 weeks). Exact session hours and timetable should be confirmed with the institute.

- Month 1 — Foundations and tool practice
- Month 2 — Intermediate topics, assignments and normal project
- Month 3 — Live project and final review

Evaluation: practical exercises, module checkpoints, code/document review where applicable, project demonstration and final feedback.

# FREE SUPPORT INCLUDED WITH EVERY COURSE

## 1 YEAR DOUBT SUPPORT + INTERVIEW PREPARATION — FREE OF COST

- One year of doubt-clearing support related to the enrolled course.
- Interview preparation support at no additional course fee.
- Guidance on explaining course projects and strengthening fundamental concepts.
- Availability, format and scheduling of support sessions to be communicated by the institute.

Important: This brochure presents the requested training offer. The institute should confirm the exact support policy, availability, start date and any applicable conditions before publishing. Interview preparation is not a job or placement guarantee.

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